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Affording Larger Brains: Testing Hypotheses of Mammalian Brain Evolution on Bats

Kate E. Jones* **and Ann M. MacLarnon†**

School of Life Sciences, University of Surrey Roehampton, West Hill, London SW15 3SN, United Kingdom

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abstract: Several major hypotheses have been proposed to explain how larger brains in mammals, such as those of humans, are afforded in energetic terms. To date, these have been largely tested on primates, with some cross-mammal analysis. We use morphological, ecological, and metabolic data for 313 species of bats to examine the allometry of brain mass and to test key predictions from three of these hypotheses: the direct metabolic constraint, expensive tissue, and maternal energy hypotheses. We confirm that megachiropteran bats (entirely fruit-eating) have larger brains for their body mass than microchiropteran bats (fruit-eating and non-fruit-eating) and fruit-eating species (Megachiroptera and Microchiroptera) have larger brains than non-fruit-eating species (Microchiroptera). Although our analyses demonstrate little or no support for any of the three hypotheses, we show that 95.9% of the variance in brain mass can be explained by the independent effects of gestation length and body mass. This indicates that among bats, the duration of maternal investment plays an important role in the adult brain mass finally obtained. These analyses serve to emphasis the crucial importance of testing the general applicability of macroevolutionary hypotheses (often developed in isolation in one clade) in multiple clades with different evolutionary histories.

Keywords: brain mass, Chiroptera, expensive tissue, independent contrasts, maternal energy investment, metabolic rate.

Following the classic work of Jerison (1973), studies of brain-size evolution have been conducted over the past 30 years on a wide range of mammalian clades (e.g., Eisenberg and Wilson 1978; Martin 1981; Gittleman 1991; Barton et al. 1995; de Winter and Oxnard 2001). Two aspects of the evolution of brain size have attracted particular attention: studies of why relatively larger brains have evolved in terms of specific selection pressures and studies of how, in energetic terms, organisms can afford to grow and maintain relatively larger brains (Martin 1998). The recent resurgence of interest in the "why" question is based on evidence that particular suites of brain structures have coevolved similarly in distantly related mammalian clades (Barton and Harvey 2000; Clark et al. 2001; de Winter and Oxnard 2001). This suggests that the sizes of different sets of brain structures respond to adaptive forces differently and thus make differential contributions to changes in overall brain size, rather than the size of different brain structures merely being the consequence of overall brain size, as suggested by Findlay and Darlington (1995). Various adaptive factors have been proposed to influence brain size or the size of specific brain structures such as foraging strategies (Eisenberg and Wilson 1978), social group sizes (Dunbar 1992), survival rates (Austad and Fischer 1992), or particular sensory systems (Jerison 1973; Barton et al. 1995; Barton 1998).

In contrast, the "how" questions have received less attention, although how organisms meet the costs of growing and maintaining this energetically expensive tissue is at least as important in understanding the factors involved in brain evolution. Investigations started from the idea that there is a metabolic constraint on growing and maintaining brain tissue. Brain tissue is expensive to maintain. For example, the mass-specific metabolic rate of the human brain is nine times higher than that of the body as a whole (Martin 1983). It is also expensive to grow because the growth of the brain is developmentally "front loaded" in mammals, occurring ahead of other organ growth (Martin 1983). Three major hypotheses have been proposed to explain how larger brains are afforded among mammal species.

Direct metabolic constraint hypothesis. Because a larger brain costs so much more in energetic terms, one might

* Corresponding author. Present address: Center for Environmental Research and Conservation, Columbia University, 1200 Amsterdam Avenue, MC 5556, New York, New York 10027; e-mail: kj2107@columbia.edu.

† E-mail: a.maclarnon@roehampton.ac.uk.

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predict that species with larger relative brain masses would have correspondingly higher basal metabolic rates (BMR). It was initially suggested that a direct relationship existed between brain mass and BMR on the basis of the similarity between the allometric scaling of brain mass and of BMR with body mass (Armstrong 1983; Hofman 1983). However, the empirical evidence suggests that only a weak direct relationship (if any) exists in mammals (Pagel and Harvey 1988*b*; McNab and Eisenberg 1989; Harvey et al. 1991; Martin 1998), although it has been suggested that if a methodological flaw is corrected in McNab and Eisenberg's (1989) analysis, there is evidence for a small positive correlation (Martin 1998). Two further hypotheses have proposed indirect metabolic mechanisms to explain how larger brains are afforded and the evolution of the consequent energetic trade-offs: the expensive tissue hypothesis (Aiello and Wheeler 1995) and the maternal energy hypothesis (Martin 1981).

Expensive tissue hypothesis. Aiello and Wheeler (1995) suggested that humans are able to maintain a large brain without a compensatory increase in BMR (relative to what would be expected from human body mass) by decreasing the amount of other metabolically expensive tissues (i.e., heart, kidneys, liver, and gastrointestinal tract) in human bodies. The expensive tissue hypothesis (ETH) builds on the direct metabolic constraint hypothesis by including other tissues as well as the brain and stating that it is essentially the metabolism of these five tissues together that must be in balance with a species BMR. Aiello and Wheeler (1995) showed that in humans and other anthropoid primates, an increase in the mass of the brain appears to be balanced by a reduction in the mass of the gastrointestinal tract. The evolutionary implication, they suggest, is that brain mass and gut mass have coevolved, at least where brain mass is large enough to pose a substantial metabolic challenge (Aiello et al. 2001). According to the ETH, oft-cited relationships between brain mass and diet, for example, the relationship between larger brains and frugivory in bats (Stephan et al. 1981) and in primates (Barton 1998), primarily reflect relationships between relative brain mass and relative gut mass, the latter being determined by dietary quality (MacLarnon et al. 1986).

In support of their hypothesis, Aiello and Wheeler (1995) found a significant negative correlation between gut mass and brain mass in anthropoid primates while controlling for the effects of body mass. Their analysis assumed that as relative gut masses and brain masses varied, the relative BMRs of species remained constant. In a later article, Aiello et al. (2001) revisited other aspects of the evidence for the expensive tissue hypothesis, showing that statistical support from analyses of anthropoid primates is very susceptible to variation in data quality and detailed analytical issues. However, they also cited increasing indications from work on birds and reptiles of the importance of energetic balance between expensive tissues (Daan et al. 1990). The expensive tissue hypothesis may provide an important explanation as to how large brain mass is maintained, but its general applicability has not been tested outside anthropoid primates.

Maternal energy hypothesis. Martin's maternal energy hypothesis (MEH; Martin 1981, 1996, 1998) suggests that the weak correlation between BMR and brain mass in adults can be explained because a direct link occurs between the energetic investment of the mammalian mother and the growth of her offspring during the periods of nutritional dependency (fetal development and infancy). Concentrating on the fetal stage, the MEH proposes that the relative BMR of the female, acting throughout the duration of the gestation period, determines the brain mass of her neonates. In precocial mammals, neonatal brain mass is on average almost half that of an adult (in altricial mammals there is a short postnatal catch-up stage to reach a similar point), so growth of the brain during gestation is a major influence on final brain mass (Martin 1983).

Some studies have reported that there is little evidence that the MEH is important in explaining the variation in the evolution of brain mass in primates (Barton 1999) and other mammals (Pagel and Harvey 1988*a*). However, as has been pointed out by Martin (1998) and MacLarnon (1999), these studies did not consider the combined effects of BMR and gestation length on neonatal brain mass. The MEH cannot be tested adequately by separate analysis of the correlation of each of the two factors with neonatal brain mass nor in multiple regression and partial correlation analyses in which other factors are simultaneously taken into account. Rather, the MEH states that the mother's BMR provides the energetic basis for the growth of fetal brains for the duration of gestation. In effect, the hypothesis states that maternal energetic investment, a variable that can be calculated as the product of gestation period and BMR, will be correlated with neonatal brain mass. Using combined residual values of gestation length and BMR, Martin (1998) found a significant positive correlation between brain mass (controlling for body mass) and this measure of maternal energy investment for a sample of 51 mammalian species from 14 orders. However, because the relationship was not significant when using the method of independent contrasts to control for the possible statistical nonindependence of these data points, the general applicability of the hypothesis remains somewhat uncertain.

We use morphological, ecological, and BMR data for 313 species of bats, a large and ecologically diverse mammalian clade, to examine how larger brains are afforded by testing the key predictions from the direct metabolic

Table 1: Phylogenetic correlation (Moran's *I*) of body mass, brain mass, basal metabolic rate (BMR), gestation length, weaning age, and intestine length within each taxonomic level

Taxonomic level	Body mass	Brain mass	BMR	Gestation length	Weaning age	Intestine length
Species with genera	.82	.86	.77	.55	.61	.61
Genera within families	.60	.74	.61	.62	.61	.73
Families within suborder	.30	.10	.13	−.36	.03	.35
Suborders within order	−.65	−.70	−.61	−.48	−.57	−.77

Note: Values of Moran's *I* vary from +1 to −1; positive values indicate that a trait at a particular phylogenetic level is more similar than random, and negative values indicate that traits are more different. All values in the table are statistically significant at the <.05 level or above. These results suggest that using species values for the five traits as independent points is statistically invalid.

constraint, expensive tissue, and maternal energy hypotheses. We carry out analyses both on species data and using an explicit phylogenetic framework. The three hypotheses described here have been proposed as fundamental principles. If so, then they should be generally applicable in different mammalian orders as well as in primates and across mammals as a whole.

Methods

Data were assembled from the literature building on our previous work on bats (Jones and MacLarnon 2001; Cruz-Neto and Jones, in press). Specifically, we collected the following variables (number of species for which we had both brain mass and trait information is given after each variable):

Adult body mass. In grams; 313 species.

Adult brain mass. In milligrams; 313 species. We use adult brain mass to test all three hypotheses. The MEH only makes direct predictions about neonatal brain mass (i.e., reflecting the amount of maternal energy invested in each litter into brain tissue). By extrapolation of the logic of MEH, brain mass at weaning would be determined by the total amount of maternal energy invested up to that point (i.e., during both gestation and lactation). In bats, infants (the majority of species have singletons) are not weaned until adult dimensions are reached (Jones and MacLarnon 2001; Jones et al. 2003). Thus, according to the MEH, bat adult brain mass should correlate with the product of BMR and gestation length plus weaning age (i.e., maternal energy investment; see below).

Adult basal metabolic rate (BMR). In milliliters of oxygen per hour, collected following the guidelines in McNab (1997; 69 species).

Total intestinal length. In millimeters; 51 species.

Gestation length. In days, excluding periods of delay, where known; 96 species.

Weaning age. In days; 86 species.

Maternal energy investment (MEI). MEI calculated as BMR × (gestation length plus weaning age); 27 species.

Diet. Fruit-eating diet includes fruit, nectar, leaves, or pollen; non-fruit-eating diet does not include fruit, nectar, or pollen; 312 species.

Taxonomy followed Koopman (1993), and values for each trait were calculated by first transforming species names in each source into their currently recognized names. Data were averaged for each of these species, combining data from different subspecies or geographic localities for all variables except diet. If there was a record of the species eating fruit, nectar, leaves, or pollen, it was recorded as fruit-eating; if there was no such record it was recorded as non-fruit-eating. Size measurements used were the average for combined samples of males and nonpregnant females. Sexual dimorphism is unlikely to bias the mass calculation, as it is rare in bats (Jones et al. 2003). Where possible we took body mass data from the same individuals used to obtain other variables. (Data file 1 and data file 2 contain data and source details. Links to these files are in the online edition of the *American Naturalist* or may be obtained from the corresponding author upon request.)

We investigated the amount of similarity in our data that was due to shared evolutionary history by using the spatial autocorrelation statistic, Moran's *I* (Gittleman and Kot 1990). Moran's *I* was used to measure the similarity of trait values at four taxonomic levels: species within genera, genera within families, families within suborders, and suborders within orders (following the taxonomy of Koopman 1993). All variables showed a significant phylogenetic autocorrelation at the level of species within genera (table 1). Because of this nonindependence, values for individual species were unlikely to represent statistically independent points in comparative analyses; if this is not taken into account, the assumptions of most statistical tests are invalidated, leading to increased Type I errors (Harvey 2000). We therefore employed independent contrasts (Felsenstein 1985), a widely used phylogenetic comparative method, to generate contrast values that attempt to control

for similarity in traits due to a shared ancestry. However, we also examined the results using species data to enable more direct comparison with previous studies and because analyses using independent contrasts have been found to have greater Type I error rates than using species values if the assumed model of evolution (Brownian motion) is incorrect (Price 1997; Harvey and Rambaut 2000; Freckleton et al. 2002).

Contrasts were generated using the CRUNCH algorithm of the CAIC computer package (Comparative Analysis by Independent Contrasts; Purvis and Rambaut 1995), and we used a species-level supertree of bats (Jones et al. 2002) to calculate the contrasts. Alternative phylogenies were not considered, although we recognize that these may be influential on results (Symonds 2002; Symonds and Elgar 2002). We found that using equal branch lengths best met the assumptions of CAIC (Purvis and Rambaut 1995), and this branch length transformation was used to standardize the independent contrasts for subsequent analyses. However, for several of the analyses, this transformation failed to standardize the contrasts adequately. To avoid inflated Type I error rates (Purvis et al. 1994), we inspected the influential contrasts in each regression analysis. Inadequately standardized contrasts are likely to be outliers from the central tendency (Purvis and Webster 1999). We therefore investigated the effect of removing these influential points on the relationships found. Influential points were recognized as having a studentized deleted residual greater than ± 3 (Jones and Purvis 1997; Jones et al. 2003). Here we report the statistics for the regressions removing influential observations, but in no case was the level of significance of those relationships affected by the inclusion of such points.

All continuous variables were $\log_e$ transformed before analysis, and statistical tests were two-tailed, performed using SPSS/PC (version 10.1). Correlations between traits were investigated using least squares linear regressions performed using both species data and independent contrasts. For independent contrasts, the least squares regression lines were constrained to go through the origin (Garland et al. 1992). Many of the analyses required the removal of the effects of body mass on one or more other variables in order to test the hypotheses adequately. In these cases, we first calculated the regression equation for each variable against body mass (using either species values or independent contrasts as appropriate) and obtained residuals from these lines, which were then used in subsequent analyses. Evidence suggests that the relationship between brain mass and body mass differs between fruit-eating and non-fruit-eating bats and also between Megachiroptera and Microchiroptera (Stephan et al. 1981; Barton et al. 1995). Note that fruit-eating species (under the definition used here) are found in both Megachiroptera and Microchiroptera, whereas non-fruit species are all microchiropterans. As these possible "grade shifts" may confound our analyses, we took a conservative approach, investigating each correlation both within all species and within the above grades, both dietary and taxonomic.

To further examine the predictions of the MEH, we took a multiple regression approach where we included as independent variables all the main effects (i.e., body mass, BMR, gestation length, and weaning age) and the interaction term (maternal energy investment), to explain the variance in adult brain mass. This approach tests for an effect of MEI that is independent of the other contributing variables. For example, a significant correlation between brain mass and MEI may be due to the effect of one of the contributing variables such as BMR or gestation length but not their combined effect. Using model simplification, we built a minimum adequate model to explain the variance in brain mass across the order and within grades. This was done by initially including all proposed variables as predictors and removing the variable responsible for the lowest reduction in variance at each step until all remaining predictors were significant ($P < .05$; following methodology in Purvis et al. 2000; Jones et al. 2003).

Results

Allometry of Brain Mass

Brain mass and body mass are highly significantly correlated in bats, using either species values or phylogenetically independent contrasts (table 2 [i]; fig. 1). The slope of the regression for all bats is different for the two methods of analyses (species values: $b = 0.81$, SE $= 0.01$, $n = 313$; independent contrasts: $b = 0.66$, SE $= 0.02$, $n = 176$). Analyzing brain mass allometry separately for Megachiroptera and Microchiroptera (see fig. 1*A* and table 2[i]; for fruit-eating and non-fruit-eating species, see fig. 1*B* and table 2[i]) suggests that the higher slope coefficient obtained for the whole sample using species values is a consequence of fitting a single allometric line across a grade shift. The values for the separate grades, by both taxonomic and dietary divisions, give more similar slope values to the one for the whole sample using independent contrasts. Analyses using independent contrasts are not affected by grade shifts in the same way as species value analyses (fig. 1*C*, 1*D*). For example, only one of the contrasts compares the change in traits between the Megachiroptera and Microchiroptera (fig. 1*C*, indicated by an asterisk). Figure 1*A*, 1*B* confirms previous suggestions that Megachiroptera and fruit-eating species have larger brain sizes for their body sizes than Microchiroptera and non-fruit-eating species, respectively, with fruit-eating versus

Table 2: Statistics for the least squares regressions (SE) of brain mass against body mass, residual basal metabolic rate (BMR), residual intestine length, and residual maternal energy investment for all species and within grades (Megachiroptera, Microchiroptera, fruit-eating, and non-fruit-eating)

	Total sample				Megachiroptera		Microchiroptera		Fruit-eating spp.		Non-fruit-eating spp.	
Hypotheses	N_T	b_T (SE)	r^2	k	N_M	b_M (SE)	N_μ	b_μ (SE)	N_F	b_F (SE)	N_{NF}	b_{NF} (SE)
i. Body mass: ln(brain mass) vs. ln(body mass)	313	.81 (.01)***	.93	3.74	52	.71 (.01)***	261	.76 (.02)***	104	.72 (.01)***	208	.73 (.02)***
	176	.66 (.02)***	.93	…	33	.62 (.02)***	143	.69 (.02)***	79	.62 (.02)***	95	.70 (.02)***
ii. Direct metabolic constraint: residual ln(brain mass) vs. residual ln(BMR)	69	.22 (.12)#	.05	.00	16	−.03 (.09)	53	.31 (.13)*	35	.04 (.08)	34	−.09 (.14)
	56	−.02 (.06)	.00	…	12	.03 (.05)	42	−.03 (.07)	31	.02 (.07)	23	−.06 (.08)
iii. Expensive tissue: residual ln(brain mass) vs. residual ln(intestine length)	51	.30 (.09)**	.18	−.08	8	.27 (.15)	43	.24 (.11)*	16	.09 (.12)	35	.10 (.11)
	40	.31 (.08)**	.26	…	5	.21 (.12)	33	.28 (.10)**	13	.17 (.14)	26	.30 (.14)*
iv. Maternal energy: residual ln(brain mass) vs. residual ln(maternal energy investment)	27	.25 (.12)#	.15	−.10	8	−.06 (.09)	19	.28 (.12)*	11	−.01 (.08)	16	.15 (.11)
	21	.04 (.06)	.02	…	7	−.00 (.04)	13	.05 (.06)	9	−.02 (.05)	10	.06 (.06)

Note: First line of statistics for each hypothesis represents results using species values; second line, results using independent contrasts. N = sample size (specifically, N_T = total, N_M = Megachiroptera, N_μ = Microchiroptera, N_F = fruit-eating , N_{NF} = non-fruit-eating), b = slope value, SE = standard error of the slope, r^2 = amount of variance explained by the regression equation, k = intercept.

$P < .1$.

* $P < .05$.

** $P < .01$.

*** $P < .001$.

non-fruit-eating producing clearer grade separation. ANCOVA results on species values demonstrate that both pairs of grade differences are significant for the scaling of brain size versus body size. Dividing the sample into suborders mega and micro, the slope values are $F = 3.01$, df = 3, 312, $P = .08$; the intercepts are $F = 3{,}652.35$, df = 3, 312, $P < .001$. Dividing the sample by dietary category, fruit and non-fruit, the slope values are $F = 0.45$, df = 3, 311, $P = .50$; the intercepts are $F = 13{,}632.52$, df = 3, 311, $P < .001$. For species analyses, residuals for relative brain mass were calculated taking this grade effect into account by calculating them from a scaling line determined by passing a line with the average of the slope values for the scaling of brain mass in the two dietary categories through the mean brain and body masses for the whole bat sample. For analyses using independent contrasts, residuals were calculated from the slope for the whole sample excluding the one contrast point with a studentized deleted residual $> \pm 3$ (the contrast between Megachiroptera and Microchiroptera) as a means of overcoming any grade effect.

Direct Metabolic Constraint

The direct metabolic constraint model predicts that relative brain mass is correlated with relative BMR, removing the effects of variation in body mass. There is no evidence of grades in the scaling of BMR to body mass across bats, and the two variables are highly correlated (species values: $b = 0.77$ [SE 0.03], $k = 0.97$, $r^2 = 0.91$, $P < .001$, $n = 96$; independent contrasts: $b = 0.78$ [SE 0.05], $r^2 = 0.81$, $P < .001$, $n = 69$; also see Cruz-Neto and Jones, in press). Thus, residuals of BMR for species analyses were calculated from the scaling slope for all bats. For species analyses, the brain mass and BMR residuals are positively correlated for both the whole sample and for Microchiroptera alone, though only the latter correlation is significant (table 2 [ii]). For the Megachiroptera and dietary categories, the species residuals are not correlated. Using independent contrasts, residual brain mass is not significantly correlated with residual BMR for bats as a whole or for any of the subsamples (table 2 [ii]). Overall, therefore, as for previous studies across mammals and in pri-

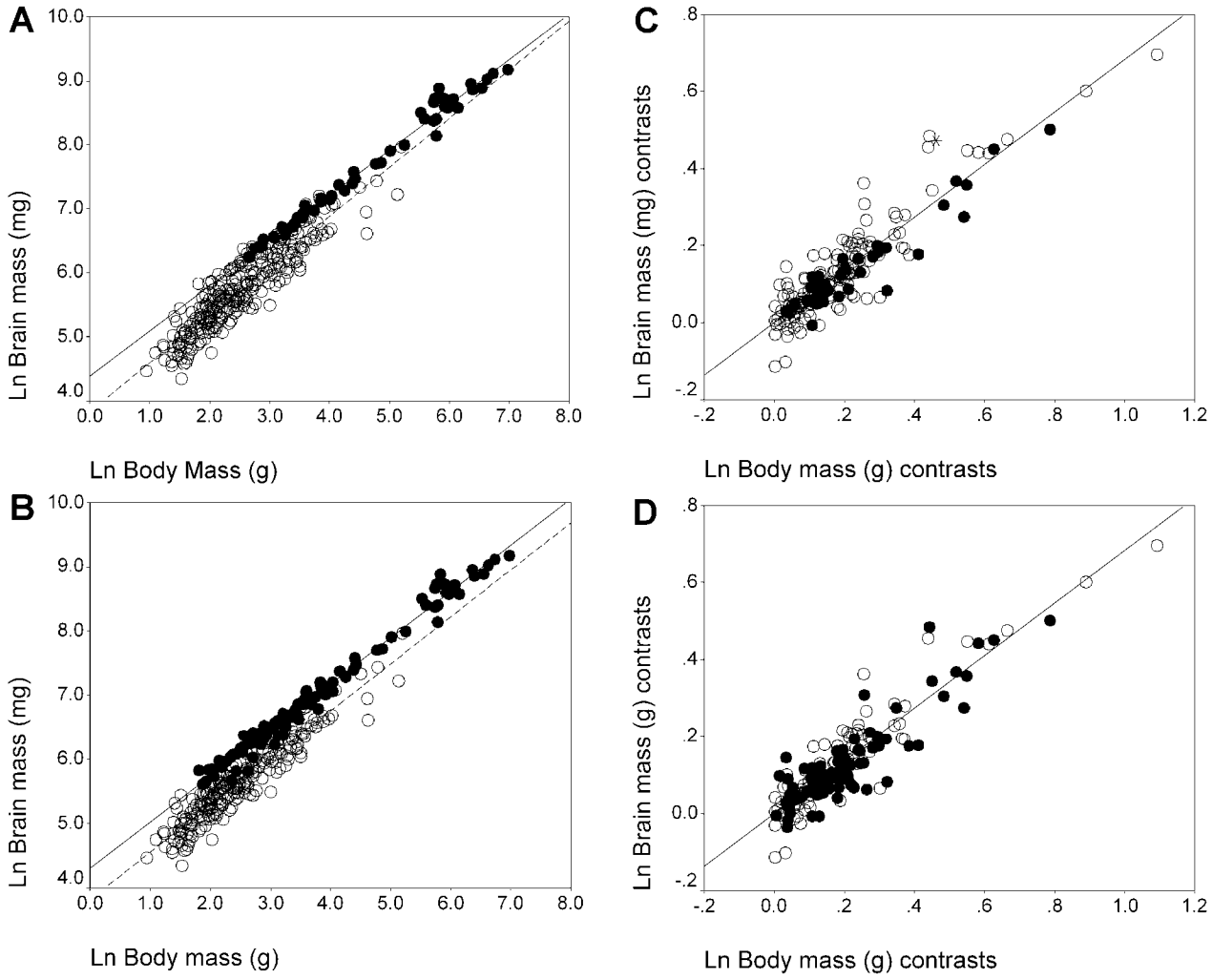


Figure 1: Scatter plots of ln(brain mass) (mg) and ln(body mass) (g) for 313 species of bats in 15 families. *A*, Species values plotted by suborder (Megachiroptera: *solid circles* and *solid line*; Microchiroptera: *open circles* and *dashed line*). *B*, Species values plotted by dietary category (fruit-eating species: *solid circles* and *solid line*; non-fruit-eating species: *open circles* and *dashed line*). *C*, Independent contrasts. Open circles are independent contrasts within Microchiroptera, and solid circles are independent contrasts within Megachiroptera; the asterisk is the contrast between Megachiroptera and Microchiroptera. Solid line = total sample less mega-micro contrast. *D*, Open circles are independent contrasts within non-fruit-eating species, and solid circles are independent contrasts within fruit-eating species. For regression statistics see table 2(i).

mates, there is very little or no support for a significant direct correlation between relative brain mass and relative BMR from bats.

Expensive Tissue Hypothesis

The ETH predicts that relative brain mass is negatively correlated with relative gut size after removing the effects of variation in body mass. There is very limited evidence of grades in the scaling of intestine length to body mass across bats (fig. 2*A*, 2*B*). Fruit bats may have slightly longer guts on average than non-fruit eaters, but because the difference is small at most, residuals for relative intestine length for species analyses were calculated from the scaling slope for all bats. For species values for the whole sample and for the largest subsample, the Microchiroptera, the residuals for relative brain mass and relative intestine length are significantly positively correlated, and for the other subsamples they are positively, if not significantly, correlated (table 2 [iii]). By independent contrasts, too, all analyses show a positive correlation between relative brain mass and relative intestine length, and this is significant across all bats, Microchiroptera and non-fruit-eating species. Both methods of analysis therefore dem-

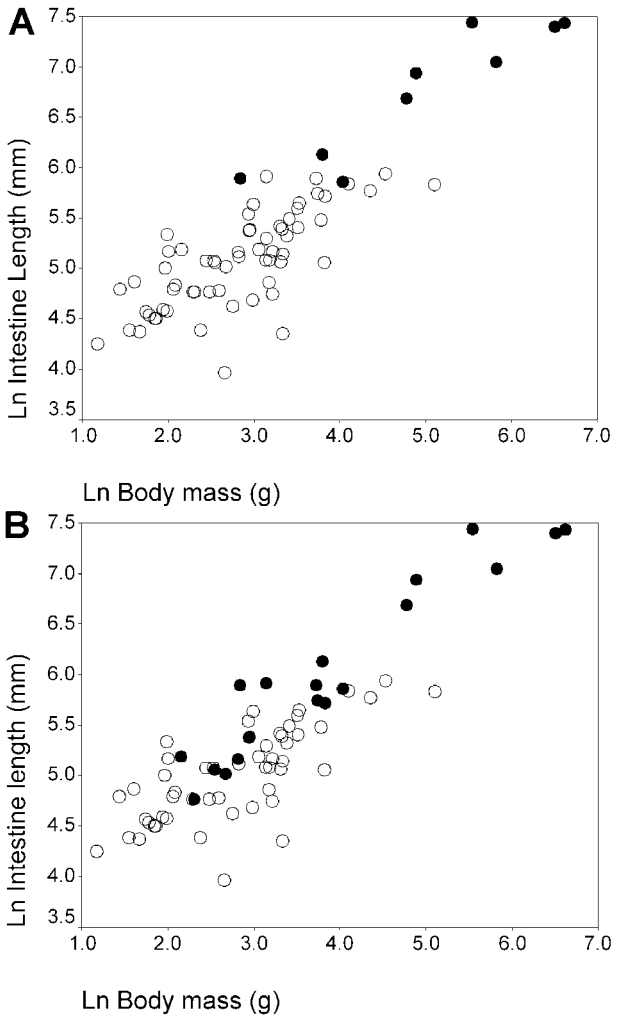


Figure 2: Scatter plots of ln(intestine length) (mm) and ln(body mass) (g; species values: ln [intestine length] = 0.58 × ln [body mass] + 3.51; contrasts: ln [intestine length] = 0.46 × ln [body mass], $n = 71$ species). *A*, Species values plotted by suborder (Megachiroptera: *solid circles*; Microchiroptera: *open circles*). *B*, Species values plotted by dietary category (fruit-eating species: *filled circles*; non-fruit-eating species: *open circles*).

onstrate that relative brain mass and relative intestine length are positively correlated in bats (table 2 [iii]), which is very different from the pattern of negative correlation found in anthropoid primates (Aiello and Wheeler 1995).

Maternal Energy Hypothesis

We test the MEH by using adult brain mass in place of neonatal brain mass and calculate maternal energy investment as the product of basal metabolic rate and the total length of maternal investment (gestation length plus weaning age). There is no evidence of grades in the scaling of maternal energy investment in bats either by suborder or by dietary category; therefore, residuals were calculated from the slope for all bats (species values: $b = 0.99$ [SE 0.06], $k = 5.20$, $r^2 = 0.89$, $P < .001$, $n = 35$; independent contrasts: $b = 0.85$ [SE 0.08], $r^2 = 0.81$, $P < .001$, $n = 25$). Species and contrast residuals for brain mass against maternal energy investment are positively correlated across bats as a whole and for the Microchiroptera and non-fruit-eating species. However, the only correlation coefficient that is significant is for species values for the Microchiroptera (table 2 [iv]). We also repeated the analysis using a second definition of maternal energy investment (the product of basal metabolic rate and gestation length, $n = 34$ species, 27 contrasts, results not shown), and none of the relationships between residuals for brain mass and residuals for maternal energy investment by this measure were significant.

The independent effects of the contributory variables for the measure of maternal energy investment used here were also tested in a multiple regression model (table 3). We find no evidence that the measure of maternal energy investment used here is independently responsible for the variance in adult brain mass when the main effect variables are also considered in the multiple regression model (table 3). The minimum adequate model for all bats using independent contrasts instead suggests that the majority of the variance in brain mass (95.9%) can be explained by two variables, body mass (partial correlation: 0.98) and gestation length (partial correlation: 0.36), with no role for the other main effect variables, BMR and weaning age. Similar models were obtained within all the subsamples (table 3) and using species values (results not shown), although gestation length was not significant in the Megachiroptera and fruit-eating subsample using species values.

Discussion

Allometry of Brain Mass

In line with previous findings, we demonstrate the presence of several grade shifts with respect to the scaling of total brain mass and body mass within the Chiroptera. Specifically, Megachiroptera (an entirely fruit-eating clade under the definition used here of fruit, pollen, leaves, and nectar) and fruit-eating species have larger total relative brain masses than Microchiroptera (a fruit-eating and non-fruit-eating clade) and non-fruit-eating species, respectively. This confirms previous results based on smaller samples and less comprehensive statistical treatment (Eisenberg and Wilson 1978; Stephan et al. 1981; Barton et al. 1995; Baron et al. 1996; de Winter and Oxnard 2001). Barton et al.'s (1995) analysis of brain mass variation in

Table 3: Independent contrasts minimum adequate model for maternal energy hypothesis variables to predict brain mass in all bats

Variable	All bats	Mega	Micro	Fruit	Non-fruit
N (contrasts)	90 (57)	17 (13)	73 (43)	26 (22)	64 (34)
r^2 (%)	95.9	98.9	94.6	98.5	95.1
Body mass	.63***	.63***	.62***	.64***	.63***
BMR	NS	NS	NS	NS	NS
Gestation length	.16**	.42**	.14*	.42**	.14*
Weaning age	NS	NS	NS	NS	NS
MEI	NS	NS	NS	NS	NS

Note: Mega = Megachiroptera, Micro = Microchiroptera, Fruit = fruit-eating species, and Non-fruit = non-fruit-eating species. Values for traits are regression coefficients (*b*). BMR = basal metabolic rate; MEI = maternal energy investment; *N* = number of species in the model with the number of contrasts in parentheses; NS = predictor not significant in the model.

* $P < .05$.

** $P < .01$.

*** $P < .001$.

65 bat species further showed that the evolutionary transition between fruit-eating and non-fruit-eating has been accompanied by an increase in the sizes of visual structures and olfactory structures in the brain, which largely accounts for the greater overall brain mass in fruit-eating bat species and Megachiroptera. They suggested that the need to identify fruit by sight and smell (i.e., sensory adaptation) may have contributed to an overall increase in brain mass rather than larger brains being an adaptation to fruit eating per se. Recent analyses of the brain architecture in 363 species of mammals also suggest that particular suites of similar-sized brain structures characterize species that have adapted to similar foraging niches that require similar sensory adaptations (de Winter and Oxnard 2001). Why brain size has evolved seems to be understood better by examining the extent to which certain brain structures have responded to particular sensory adaptations and the contribution of such changes to total brain mass rather than looking for explanatory correlates of overall brain mass (Barton and Harvey 2000).

Direct Metabolic Constraint Hypothesis. There is only very limited or no support from bats for the hypothesis of a direct metabolic constraint on brain growth and maintenance. First, there is no clear similarity of scaling between brain and body mass and between BMR and body mass. Second, brain mass and BMR are only significantly correlated by one method within one suborder when controlling for body mass. Third, there was no evidence from the minimum adequate multiple regression model that BMR played any significant role in explaining the variance in brain mass across all bats or within grades. Martin (1998) found a weak significant relationship between relative brain mass and relative BMR across mammals as a whole. Barton (1999) also presented evidence for a direct link between BMR and brain mass in another clade of mammals, the primates. However, further analysis within primate grades and using independent contrasts to control for phylogenetic inertia demonstrated no evidence for a significant direct relationship (Barton 1999). Overall, the evidence is mounting that BMR is neither solely nor directly responsible for variance in brain mass.

Expensive Tissue Hypothesis. Aiello and Wheeler's (1995) expensive tissue hypothesis offers a mechanism whereby organisms can afford to maintain larger brains without an accompanying increase in metabolic rate by reducing other metabolically expensive tissues in the body. In bats, however, relative brain mass and relative intestine length, a measure of the size of one of the most important other expensive tissues, are positively correlated, contradicting the hypothesis. The evolution of a larger brain in bats has not been accompanied by a reduction in intestine size as has been found in anthropoid primates (Aiello and Wheeler 1995). A significant positive correlation between brain mass and intestine length in bats is found regardless of analysis type when all species are considered, though it is only found by one analysis type within the Microchiropteran and non-fruit-eating species grades. However, although low sample sizes may have reduced our power to detect significant relationships within grades, there is certainly no indication that these variables are negatively correlated, as predicted.

These results suggest that the ETH is not upheld in bats and hence is not generally applicable. Closer examination shows that the anthropoid sample used by Aiello and Wheeler (1995) essentially comprised only frugivores and folivores. In outline, their results indicate that frugivorous

primates have larger brains and smaller intestines than folivorous species. Results from bats indicate that frugivorous species have larger brains and possibly slightly larger intestines than non-fruit-eating species, most of which are insectivorous. Because the associations between brain mass and intestine size are so different in the two orders, this suggests that they are not causally linked. Rather, it suggests that different factors are operating on each of the two variables in relation to diet. Among mammals, insectivores generally have somewhat smaller intestines relative to body mass than frugivores, which have smaller intestines than folivores, reflecting the relative digestibility of the different foodstuffs (MacLarnon et al. 1986). However, relative brain mass does not consistently follow the same pattern across these three dietary categories as exemplified by the results presented here. Hence, relative brain mass and relative intestine size would not be expected to be correlated across the three dietary categories or similarly correlated across different pairs of dietary categories as is found in the analyses presented here.

It is possible that another expensive tissue is confounding the picture in bats. Heart mass in bats is about 1.5–2.0 times larger than in other mammals relative to body mass, which is associated with the energetic costs of flight, as in birds (Schmidt-Nielsen 1984). The energetic cost of heart tissue is therefore of greater importance in maintaining energy balance in bats than it is in other mammals. However, there is no evidence that relative heart mass varies significantly within bats (see fig. 3), and their relative heart mass is not correlated with relative brain mass (species values: $r^2 = 0.01$, $P = .74$, $n = 19$; data from Cruz-Neto and Jones, in press). Hence, variation in heart mass is unlikely to be acting as a confounding factor in analyses of other expensive tissues within the order.

Testing the expensive tissue hypothesis by comparing the relative sizes of two tissues requires the assumption that variation in BMR is not influential in the sample under consideration. On average, bats have lower BMRs than other mammals (fig. 4). However, they do not have particularly variable BMRs among themselves compared with other mammalian orders, and BMR is highly correlated with body mass in bats. This variation in BMR is unlikely to be acting as a confounding factor in examining the energetic balance between expensive tissues within bats. This can be tested directly by multiple regression analyses of brain mass, body mass, intestine length, and BMR. Using both species data and independent contrasts, the results again show a significant positive correlation between brain mass and intestine length when the effects of both BMR and body mass are taken into account simultaneously (species values: $r^2 = 0.95$, $P =< .01$; independent contrasts: $r^2 = 0.96$, $P =< .05$).

One further possibility is that the smaller brain masses

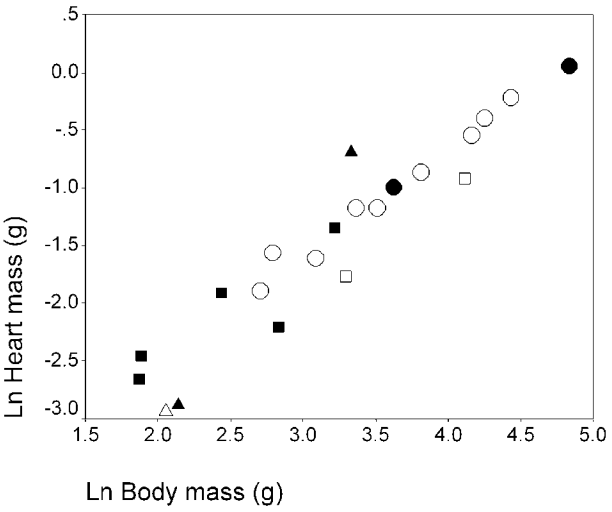


Figure 3: Scatter plot of ln(heart mass) (g) and ln(body mass) (g) for 21 species of bats in six families using species values taken from Cruz-Neto and Jones (in press) (ln [heart mass] = 0.91 × ln [body mass] − 4.36). Solid circles = Pteropodidae, open circles = Phyllostomidae, solid triangles = Rhinolophidae, open triangles = Emballonuridae, solid squares = Vespertilionidae, open squares = Noctilionidae. Circles represent fruit-eating species, and squares and triangles represent non-fruit-eating species.

of bats, in contrast to anthropoid primates, do not pose a sufficient metabolic challenge to require significant compensation in the mass of other expensive tissues such as gut size. However, this would be expected to result in no correlation between relative brain mass and relative gut length, not the positive correlation found. Overall, the results presented here from bats certainly challenge the general macroevolutionary applicability of the ETH as an explanation for the principles underlying mammalian brain evolution. They also throw doubt on whether there is a causal basis for the correlation previously found between relative brain mass and relative gut size in anthropoid primates (Aiello and Wheeler 1995).

Maternal Energy Hypothesis. The evidence from bats offers no support for the hypothesis that the combined effects of maternal BMR and the length of the maternal investment period are important in determining the growth of the brain. However, our analysis using minimum adequate models suggests that at least in bats, the length of the gestation period plays a more important role in the final adult brain mass. For example, 95.9% of the variance in brain mass in bats can be explained by the independent effects of gestation length and body mass. To grow a larger brain, it seems that bats need to gestate their infants for longer and that variation in basal metabolism has no im-

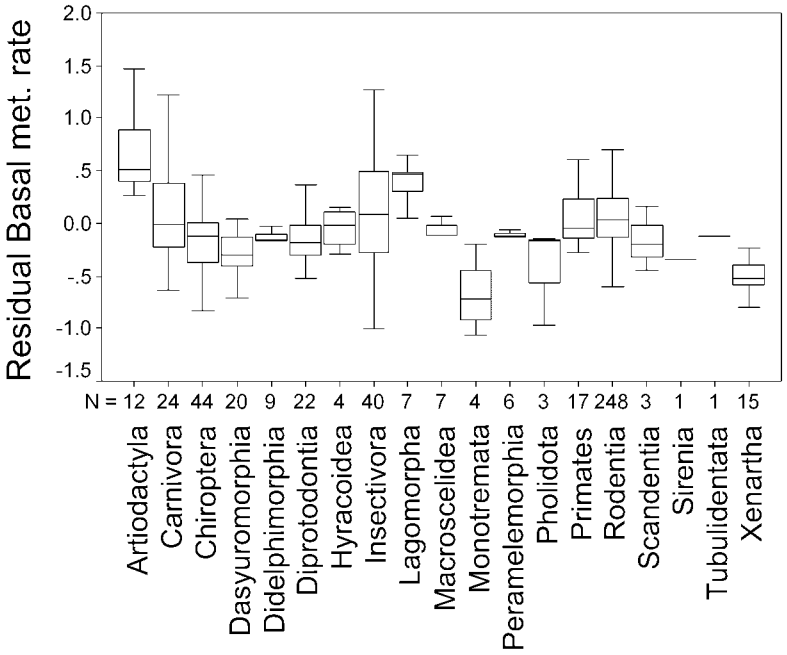


Figure 4: Box plot for residual ln(BMR) (mL O_2/hr) for 487 mammalian species in 19 orders taken from Lovegrove (2000). Residuals were obtained from the allometric slope of the whole sample using species values (ln [BMR] = 0.69 × ln [body mass] + 1.41).

pact. These results are in contrast to those obtained from Martin's (1998) analysis across 51 mammalian species from 14 orders that found evidence for a significant relationship of neonatal brain mass and maternal energy investment (controlling for maternal mass). However, this result was obtained only using species as independent points but not when independent contrasts were used to correct for possible nonindependence in the species data. When we repeated our analysis using MEI as the single predictor of brain mass (controlling for body mass), we found only one significant correlation (within Microchiroptera using species values) to support the MEH out of a total of 11 (across bats, within grades and types of analysis). This suggests that the results within bats are even robust to different methodologies.

Our findings within bats probably have wider applicability within mammals in general. For example, the bat analyses are consistent with a much earlier study that found that gestation length correlates very highly with neonatal brain mass after controlling for maternal mass in 116 mammal species (using species as independent points; Pagel and Harvey 1988*a*). The relationship between brain mass, length of the period of maternal investment, and metabolism remains to be explored further with explicitly independent data and within different grades (such as the developmental state of the infant at birth) and clades. For example, an analysis within primates (Barton 1999) showed that adult brain mass is not correlated with gestation length independently of metabolic rate, although this was based on small sample sizes ($n = 18$ independent contrasts), suggesting that these variables may interact differently in different groups. What the present results indicate, though, is that in bats, variation in gestation period affects brain size but that variation in BMR does not. Hence the maternal energy hypothesis, which involves the combination of gestation period and BMR, is not supported independently of the effect of gestation period alone.

Conclusions

None of the three major hypotheses tested here that have been proposed to explain how mammals afford larger brain sizes have any substantial support from bats. The direct metabolic constraint hypothesis and the maternal energy hypothesis (sensu stricto) are not supported, and the expensive tissue hypothesis is contradicted. However, gestation does play an important role in the adult brain mass finally obtained in bats, and hence maternal investment by some measure is important. It remains possible that different factors contributing to maternal investment may be important in different clades, but results from those investigated to date indicate that variation in gestation period is a common factor. Whether gestation pe-

riod determines brain mass or brain mass determines gestation period (Sacher and Staffeldt 1974) cannot be uncovered by correlation tests alone, and the proximate physiological mechanisms underlying the association found are unclear. These analyses serve to emphasize the crucial importance of testing the general applicability of macroevolutionary hypotheses (often developed in isolation in one clade) in multiple clades with different evolutionary histories.

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